

MIET2572 – Wearable Systems Data Lab

Week 5-6: From Signal to System

Dr Francisco Tovar

Objective

In this lab you will analyse how a wearable system produces heart rate from physiological activity.

Key idea: A wearable does not measure heart rate. It infers it from a signal and delivers it through a system.

System Overview

Body → Sensor → Processing → Output

A wearable system is not a sensor. It is a chain of dependencies.

Task 1 – Observe the System (HR Band)

Action

- Observe heart rate measurements from the Polar HR band
- Record values at rest and after exercise

Analyse

- Does the value change instantly?
- Is the measurement stable or delayed?

Question

- Where does this number come from?
-

Task 2 – Estimate Heart Rate from Observation

Action

- Count your pulse manually for 15 seconds
- Convert to beats per minute (bpm)
- Compare with the HR band reading

Analyse

- Do both measurements match?
- Which one responds faster to changes?

Question

- Is heart rate directly measured or inferred?
-

Task 3 – System Response

Action

- Perform a sudden activity (e.g. squats)
- Observe how HR changes over time

Analyse

- Does HR change instantly or gradually?

Question

- What does this suggest about internal processing?
-

Task 4 – Model the System

Action

- Draw a block diagram with exactly 3 boxes and 2 arrows

Rules

- Box 1 = what you change (input)
- Box 3 = what you observe (output: HR)
- Box 2 = unknown internal process

Constraint

- You must label Box 2 using ONLY observations from the lab

Evidence

- Write one sentence explaining:
 - why HR does not change instantly

Question

- What does Box 2 do?

Task 5 – System Sensitivity

Action

- Loosen the sensor slightly
- Measure HR at rest and during movement

Record

- HR (stable condition)
- HR (loose / motion condition)

Analyse

- Does HR become unstable, delayed, or incorrect?

Question

- What type of error is introduced (noise, delay, or bias)?
-

Task 6 – Dynamic Response (HR vs Time)

A change in activity produces a transient response in heart rate.

The system does not jump instantly to a new value. It evolves over time until it stabilises. We characterise this behaviour using a time constant τ , which represents how fast the system responds.

Action

- Record HR as fast as the device updates
 - 30 s rest
 - Sudden activity
 - 60 s recovery
- (Optional) Plot HR vs time

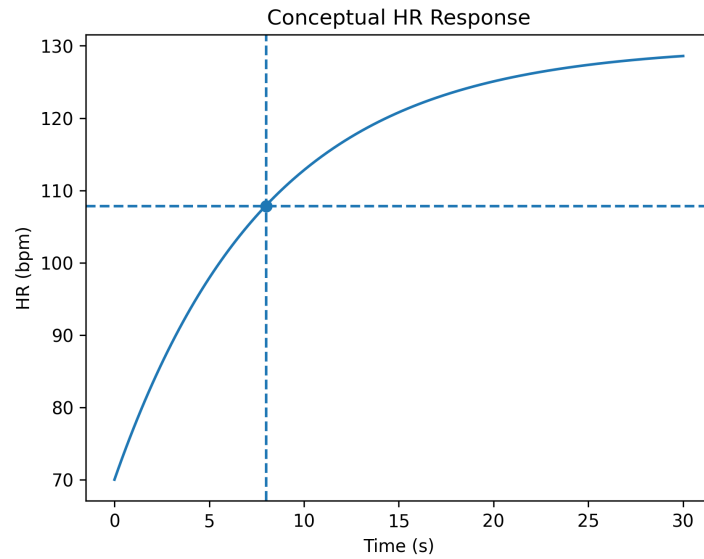
Guideline

If the system were instantaneous, HR would jump immediately.

Instead:

filtering $\Rightarrow$ dynamics $\Rightarrow \tau$

Expected Behaviour



Estimate

- HR_0 = initial value
- HR_f = final value
- $\Delta HR = HR_f - HR_0$
- $HR_\tau = HR_0 + 0.63 \cdot \Delta HR$
- Identify the first time at which HR reaches HR_τ

You don't need a perfect curve. You just need to find when it gets most of the way there

Record

Time (s)	HR (bpm)
0	
5	
10	
...	

Comparison

- Repeat with tight vs loose sensor
- Compare τ

Error Estimate

- Assume uncertainty ± 5 s

Question

- What does τ represent in this system?

Class Discussion

Compare your τ with other groups. Are the results consistent?

System-Level Reflection

Action

- Identify where delay and error are introduced

Analyse

- Is delay due to the body or the device?
- Which part do you trust least?

Results Log

Condition	HR_0	HR_f	τ (s)
Tight			
Loose			

Final Reflection

- What defines a “heartbeat”?
- Where can the system fail?
- Is the delay due to the body or the device?
- How would you redesign the system to reduce τ ?

Key Takeaway

Signals $\rightarrow$ Models $\rightarrow$ Systems $\rightarrow$ Interpretation

The challenge is not collecting data. The challenge is trusting it.